

CE 536 – Introduction to Numerical Methods for Civil Engineers

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Course Outline

- MATLAB Basics (2 weeks)
 - General MATLAB commands.
 - Simple examples.
- Numerical integration and differentiation (1.5 wks)
 - Simple engineering applications.
- Ordinary differential equations (2 wks)
 - Runge-Kutta methods.
 - Applications in structural and environmental engineering.
- Partial differential equations (3 wks)
 - Finite-difference and finite-element methods.
 - Linear and non-linear system solution (direct and iterative methods).
 - Applications in groundwater flow and transport.

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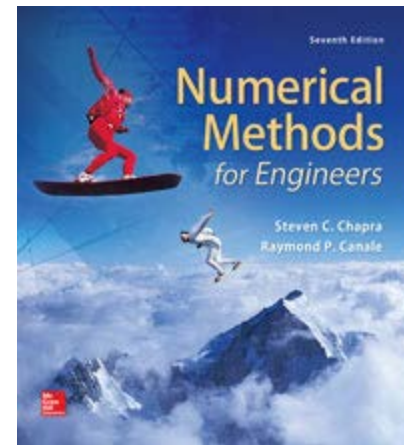
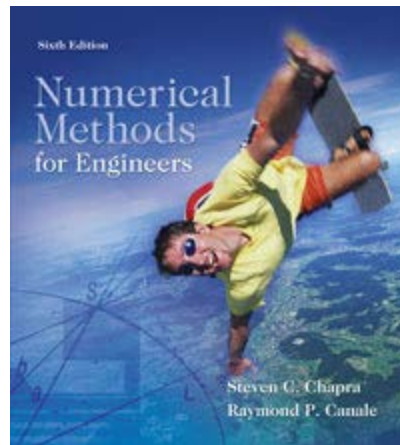
- Optimization (2 weeks)
 - Classical methods, Gradient-based methods, Heuristic methods
 - Design applications in structural and environmental engineering.
- Curve fitting and inverse problems (1.5 weeks)
 - Curve fitting (linear and non-linear regression)
 - Direct methods for inverse problems
 - Indirect methods for inverse problems
 - Applications in water resources engineering
- Other MATLAB features (0.5 week)

Grading etc.

- Assignments (35%)
 - 7 assignments
 - Due 1.5 weeks from assignment
- Project Assignment (20%)
 - Individual
 - Assigned after spring break
 - Due on last day of class
- Quizzes (10%)
- Final exam (35%) – 10 days take home
 - About 10-12 questions
- Grading
 - A weighted grade of 97 or above is guaranteed a course grade of A+, 90-96 = A/A-, 78 - 89 = B-/B/B+, 65-77 = C-/C/C+, and 50-64 = D-/D/D+.

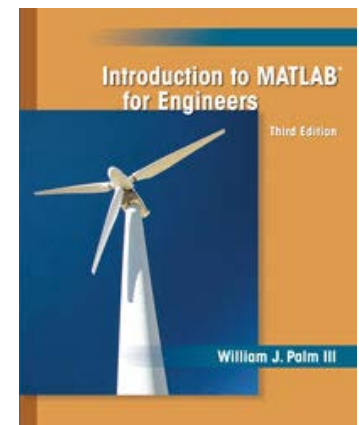
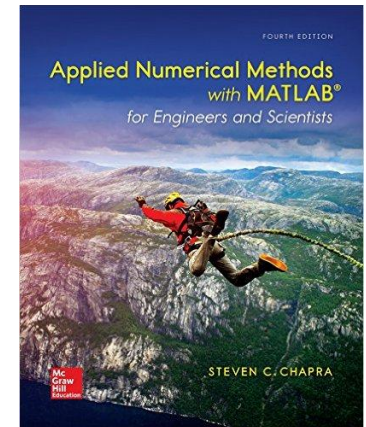
Text Book

- Chapra, S.C., and, R.P. Canale, **Numerical methods for engineers**, McGraw Hill, 5th/6th/7th Edition, 2006/2010/2015. (CC)



Other Recommended Texts

- Chapra, S.C., **Applied Numerical methods with MATLAB**, 4th edition, McGraw Hill, 2018. (Chapra-18)
 - Many CE applications
 - **Strongly Recommended text**
- Palm W.J. III, **Introduction to MATLAB for engineers**, 3rd edition, McGraw Hill, 2010. (Palm)
 - Chapra-Canale & Palm or Chapra & Palm can be bought as a package from McGraw Hill for a lower price.



Some Additional Texts

- Schilling, R.J., and S.L. Harris, **Applied numerical methods for engineers using MATLAB and C.**
 - Good applications and MATLAB codes
- Borse, G.J., **Numerical methods with MATLAB**, PWS publishing, 1997.
 - Mostly methods – not much apps
 - some good MATLAB examples.
- **Exploring Numerical Methods: An introduction to scientific computing using MATLAB**, Linz and Wang, Johns and Bartlett Publishers, 2003.
 - Similar to Borse
- Hoffman, J.D., **Numerical methods for Scientists and Engineers**, McGraw Hill, 1992.
 - Good coverage of PDE methods – not much focus on applications.

MATLAB Help

- Tutorial:
 - Onramp Tutorial (2 hours):
<https://www.mathworks.com/training-schedule/matlab-onramp.html>
 - Academic Tutorial (6+ hours):
<https://matlabacademy.mathworks.com>
- Documentation:
 - <http://www.mathworks.com/help/>